Geometry Similarity Cheat Sheet

Unit Focus: *scale factors, similar figures, and proportions*

1 KEY VOCABULARY

similar figures	Figures that have the **same shape** but not necessarily the **same size**.
corresponding angles	Angles in the same relative position that are **congruent** (equal in measure).
corresponding side lengths	Side lengths in the same relative position that are **proportional**.
scale factor	The number that multiplies the original figure to get the image. If the scale factor is k, then each corresponding side length is multiplied by k.

2 SIMILARITY STATEMENT

If polygon $ABCD$ is similar to polygon $EFGH$, then the order of the vertices tells which parts correspond.

$$ABCD \sim EFGH$$



Corresponding parts:

$\angle A \cong \angle E,\ \angle B \cong \angle F,\ \angle C \cong \angle G,\ \angle D \cong \angle H$

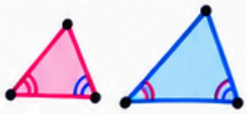
$\overline{AB} \leftrightarrow \overline{EF},\ \overline{BC} \leftrightarrow \overline{FG},\ \overline{CD} \leftrightarrow \overline{GH},\ \overline{DA} \leftrightarrow \overline{HE}$

3 TRIANGLE SIMILARITY CRITERIA

If two triangles meet any of the following criteria, they are similar.

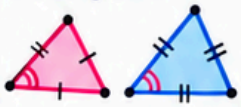
AA (Angle-Angle)

If two **angles** of one triangle are congruent to two angles of another triangle.



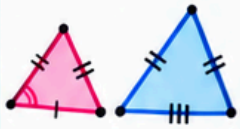
SAS ~

If two pairs of corresponding sides are proportional and the included angles are congruent.



SSS ~

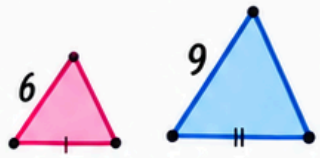
If all three pairs of corresponding sides are proportional.



4 FINDING A SCALE FACTOR

Compare a side length from the original figure to the corresponding side length in the image.

Example: A side length changes from 6 to 9. Find the scale factor from the original to the image.



Original (smaller) Image (larger)

Step 1: Write a **ratio** (image ÷ original).

$$\frac{9}{6}$$

Step 2: Simplify.

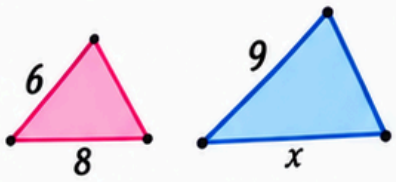
$$\frac{9}{6} = \frac{3}{2}$$

Answer: $\frac{3}{2}$

5 FINDING A MISSING SIDE

Use a proportion with corresponding side lengths.

Example: In similar triangles, $\frac{6}{9} = \frac{8}{x}$. Find x.



Step 1: Set up a proportion.

$$\frac{6}{9} = \frac{8}{x}$$

Step 2: Cross multiply.

$$6x = 72$$

Step 3: Solve.

$$x = 12$$

Answer: $x = 12$

6 PERIMETER AND SIMILAR FIGURES

The ratio of the perimeters of two similar figures equals the scale factor.

Example: If the scale factor from a smaller figure to a larger figure is 2, what is the ratio of their perimeters?



Original $P = 10$ Image (scale factor 2) $P = 20$

Perimeter ratio (image : original)

= scale factor

= 2 : 1

Answer: 2 : 1

7 AREA AND SIMILAR FIGURES

The ratio of the areas of two similar figures equals the square of the scale factor.

Example: If the scale factor from a smaller figure to a larger figure is 3, what is the ratio of their areas?



Original $A = 4$ Image (scale factor 3) $A = 36$

Area ratio (image : original)

$= (\text{scale factor})^2$

$= 3^2$

$= 9 : 1$

Answer: 9 : 1

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Use matching (corresponding) sides when setting up proportions.
- Keep the same order (small to large or large to small) on both ratios.
- The scale factor multiplies all corresponding side lengths.
- Perimeters are in the same ratio as the scale factor.
- Areas are in the ratio of the square of the scale factor.

COMMON MISTAKES:

- Mixing up the order of corresponding sides in a proportion.
- Using non-matching sides that do not correspond.
- Forgetting to square the scale factor when finding area ratios.
- Applying the scale factor to area instead of squaring it.
- Assuming figures are similar without checking the given information or a similarity criterion.

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Geometry Similarity Cheat Sheet

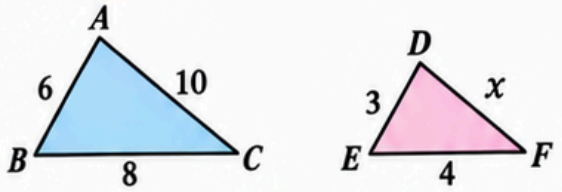
More Practice: *similar triangles, dilations, and applications*

1 USING PROPORTIONS IN SIMILAR TRIANGLES

If two triangles are similar, set corresponding side ratios equal.

If $\triangle ABC \sim \triangle DEF$, then $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$.

Example: $\triangle ABC \sim \triangle DEF$. Find x.



$\frac{6}{3} = \frac{8}{4} = \frac{10}{x}$ So $\frac{10}{x} = \frac{6}{3} = 2 \Rightarrow x = 5$

Answer: $x = 5$

2 AA SIMILARITY EXAMPLE

If angle A = angle D and angle B = angle E, then triangle ABC is similar to triangle DEF by AA.

Corresponding sides: AB / DE = BC / EF.

Example: AB = 6, DE = 4, BC = 9.

Find EF = x.

6 / 4 = 9 / x => 6x = 36

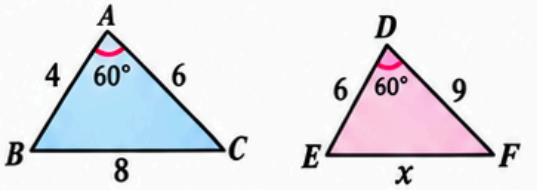
Answer: x = 6.

3 SAS~ EXAMPLE (SIDE-ANGLE-SIDE)

If the included angle is congruent and the surrounding sides are proportional, then the triangles are similar.

If $\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle A \cong \angle D$, then $\triangle ABC \sim \triangle DEF$.

Example: The marked angles are congruent. Find x.



$\frac{4}{6} = \frac{8}{x}$

$4x = 48$

$x = 12$

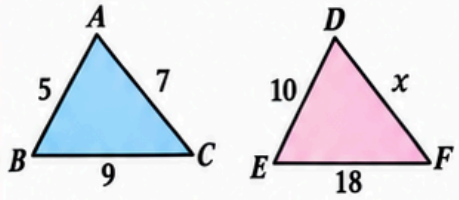
Answer: $x = 12$

4 SSS~ EXAMPLE (SIDE-SIDE-SIDE)

If all three pairs of sides are proportional, then the triangles are similar.

If $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$, then $\triangle ABC \sim \triangle DEF$.

Example: The triangles are similar. Find x.



$\frac{5}{10} = \frac{9}{18} = \frac{7}{x}$

Use $\frac{5}{10} = \frac{7}{x}$.

$5x = 70 \Rightarrow x = 14$

Answer: $x = 14$

5 DILATIONS

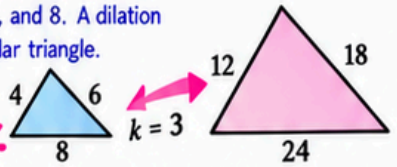
A dilation with scale factor k multiplies every length by k and preserves angle measures.

- If $k > 1$, the figure is an enlargement.
- If $0 < k < 1$, the figure is a reduction.

Example: A triangle has sides 4, 6, and 8. A dilation with scale factor $k = 3$ creates a similar triangle. Find the new side lengths.

Multiply each side by 3:

$4 \rightarrow 12$, $6 \rightarrow 18$, $8 \rightarrow 24$

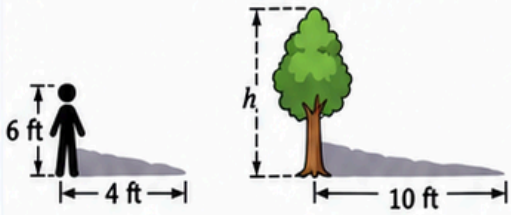


Answer: 12, 18, 24

6 INDIRECT MEASUREMENT EXAMPLE

Use similar triangles to find an unknown height.

Example: A 6-foot person casts a 4-foot shadow. A tree casts a 10-foot shadow. Find the height h of the tree.



Use similar triangles:

$\frac{6}{4} = \frac{h}{10}$

$4h = 60$

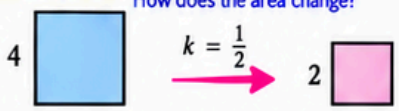
$h = 15$

Answer: $h = 15$ feet

7 RATE OF CHANGE OF AREA UNDER DILATION

If a figure is dilated by a scale factor k, all lengths are multiplied by k, and the area is multiplied by k^2.

Example: A square is dilated by a scale factor of $k = \frac{1}{2}$. How does the area change?



Area $= 4^2 = 16$

Area $= 2^2 = 4$

The area is multiplied by $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$, so the new area is $\frac{1}{4}$ of the original area.

8 QUICK STRATEGY TIPS & COMMON MISTAKES

STRATEGY TIPS:

- Keep corresponding sides in the same order when setting up proportions.
- Make sure the triangles are similar (AA, SAS~, or SSS~) before using proportions.
- Use the same scale factor for all lengths.
- Remember: areas scale by k^2, not k.
- Label your diagrams and mark corresponding angles and sides.
- Check your answer by seeing if it makes sense in the context (e.g., reasonable length or height).

COMMON MISTAKES:

- Matching the wrong sides in proportions (not keeping the same order).
- Assuming triangles are similar without enough information.
- Using the linear scale factor k when you need the area ratio k^2 (or vice versa).
- Forgetting to square the scale factor for area.
- Arithmetic errors when setting up or solving proportions.

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